

Data Science – Comprehensive Syllabus

Course Overview

This course provides a complete end-to-end learning path in Data Science, combining programming, statistics, data analysis, machine learning, and model deployment. Students will learn how to extract insights from data and build intelligent predictive systems using real-world datasets.

Duration: 12–16 Weeks (80–110 Hours)

Prerequisites:

- Basic Python Programming
 - Basic Mathematics (Statistics & Algebra recommended)
 - Basic understanding of data handling
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Module 1: Introduction to Data Science (Week 1)

Topics

- What is Data Science?
- Data Science vs Data Analytics vs AI/ML
- Data Science Lifecycle
- Roles in Data Science
- Industry Applications
- Tools & Technologies Overview

Practical Exercises

- Installing Python data science stack
 - Exploring sample datasets
 - Basic data inspection
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Module 2: Python for Data Science (Week 2)

Topics

- Python refresher for data science
- NumPy fundamentals
- Pandas for data manipulation
- Data loading and exporting
- Handling missing data

Practical Exercises

- Dataset loading and cleaning
 - Data filtering and grouping
 - Statistical summaries using Pandas
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Module 3: Data Collection & Data Wrangling (Week 3)

Topics

- Data Sources (CSV, JSON, APIs, Databases)
- Data Cleaning Techniques
- Handling Missing Values
- Removing Duplicates
- Data Transformation
- Feature Formatting

Practical Exercises

- Cleaning real-world messy datasets
 - API-based data collection
 - Data preprocessing pipeline
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Module 4: Exploratory Data Analysis (EDA) (Week 4)

Topics

- Data Understanding Techniques
- Summary Statistics
- Correlation Analysis
- Distribution Analysis
- Outlier Detection
- Feature Relationships

Practical Exercises

- Titanic dataset EDA
 - Sales dataset analysis
 - Customer behavior insights
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Module 5: Data Visualization & Storytelling (Week 5)

Topics

- Matplotlib Basics
- Seaborn Advanced Visualization
- Types of Charts (Bar, Line, Scatter, Heatmap)
- Dashboard Basics
- Data Storytelling Techniques

Practical Exercises

- Business dashboard creation
 - Trend analysis charts
 - Insight reporting
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Module 6: Statistics for Data Science (Week 6)

Topics

- Descriptive Statistics
- Probability Concepts
- Normal Distribution
- Hypothesis Testing
- Correlation vs Causation
- Confidence Intervals

Practical Exercises

- Statistical analysis using Python
 - Hypothesis testing examples
 - Data interpretation tasks
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Module 7: Machine Learning Fundamentals (Week 7)

Topics

- Introduction to Machine Learning
- Types of Machine Learning
- Supervised vs Unsupervised Learning
- Model Training Process
- Train-Test Split

Practical Exercises

- First ML model creation
 - Dataset training and testing
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Module 8: Supervised Learning (Week 8)

Topics

- Linear Regression
- Multiple Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- KNN Algorithm

Practical Exercises

- House price prediction
 - Customer churn prediction
 - Classification models
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Module 9: Model Evaluation & Optimization (Week 9)

Topics

- Accuracy, Precision, Recall, F1 Score
- Confusion Matrix
- ROC-AUC Curve
- Cross Validation
- Overfitting & Underfitting
- Hyperparameter Tuning

Practical Exercises

- Model comparison experiments
 - Accuracy improvement tasks
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Module 10: Unsupervised Learning (Week 10)

Topics

- Clustering Concepts
- K-Means Clustering
- Hierarchical Clustering

- PCA (Principal Component Analysis)
- Dimensionality Reduction

Practical Exercises

- Customer segmentation
 - Market clustering
 - PCA visualization
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Module 11: Introduction to Deep Learning (Week 11)

Topics

- Neural Networks Basics
- Perceptron Model
- Activation Functions
- Forward & Backpropagation
- TensorFlow / Keras Basics

Practical Exercises

- Simple neural network model
 - MNIST digit classification
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Module 12: Natural Language Processing (NLP) (Week 12)

Topics

- Text Processing Techniques
- Tokenization
- Bag of Words
- TF-IDF
- Sentiment Analysis
- Text Classification

Practical Exercises

- Sentiment analysis model
 - Text classifier development
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Module 13: Model Deployment (Week 13)

Topics

- Model Saving (Pickle/Joblib)
- Flask / FastAPI for ML
- API Creation for Models
- Introduction to Cloud Deployment
- Streamlit Basics

Practical Exercises

- Deploy ML model as API
 - Build prediction web app
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Module 14: Mini Project (Week 14)

Choose one:

- House Price Prediction System
 - Customer Segmentation System
 - Spam Email Classifier
 - Movie Recommendation System
 - Disease Prediction System
 - Sales Forecasting System
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Module 15–16: Capstone Project

End-to-End Data Science System

- Problem Definition
- Data Collection
- Data Cleaning & Preprocessing
- Exploratory Data Analysis

- Feature Engineering
 - Model Building
 - Model Evaluation
 - Hyperparameter Tuning
 - Deployment (Web/API)
 - Documentation
 - Final Presentation
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Tools & Technologies

- Python
 - NumPy, Pandas
 - Matplotlib, Seaborn
 - Scikit-learn
 - TensorFlow / Keras
 - Jupyter Notebook / Google Colab
 - Flask / FastAPI
 - Streamlit
 - SQL (MySQL / PostgreSQL)
 - Git & GitHub
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Core Concepts Covered

- Data Science Lifecycle
 - Data Wrangling & Cleaning
 - Exploratory Data Analysis (EDA)
 - Statistical Analysis
 - Data Visualization & Storytelling
 - Machine Learning Algorithms
 - Model Evaluation & Optimization
 - Unsupervised Learning
 - Deep Learning Basics
 - Natural Language Processing (NLP)
 - Model Deployment
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Assessment Scheme

Component	Weightage
Assignments	15%
Lab Exercises	25%
Quizzes	10%
Mini Project	20%
Capstone Project	30%

Learning Outcomes

By the end of this course, learners will be able to:

- Understand the complete Data Science lifecycle from raw data to deployment.
- Clean, transform, and analyze large datasets using Python tools.
- Perform exploratory data analysis and extract meaningful insights.
- Apply statistical methods for data-driven decision making.
- Build supervised and unsupervised machine learning models.
- Develop basic deep learning and NLP applications.
- Evaluate and optimize machine learning models effectively.
- Deploy predictive models using APIs and web frameworks.
- Work on real-world datasets across industries like healthcare, finance, retail, and technology.
- Build complete end-to-end Data Science projects ready for production.